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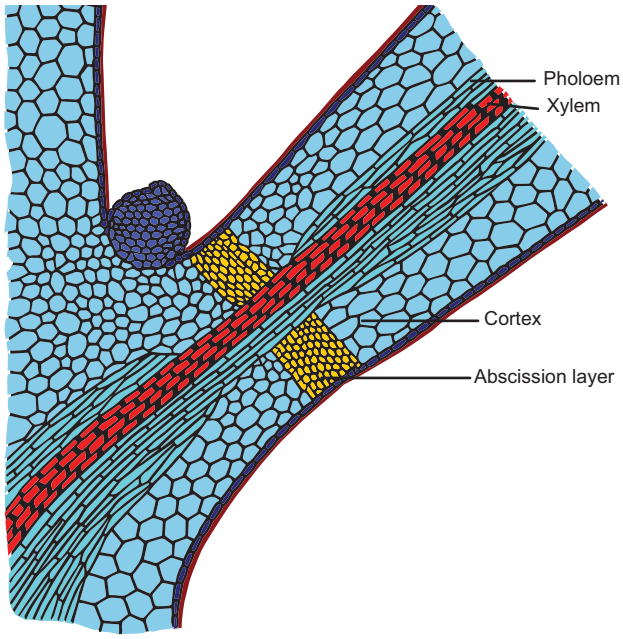


Figure 15.30: L.S of petiolar base showing abscission layer

7. Hormones influencing abscission

All naturally occurring hormones influence the process of abscission. Auxins and cytokinins retard abscission, while abscisic acid (ABA) and ethylene induce it.

8. Significance of abscission

1. Abscission separates dead parts of the plant, like old leaves and ripe fruits.
2. It helps in dispersal of fruits and continuing the life cycle of the plant.
3. Abscission of leaves in deciduous plants helps in water conservation during summer.
4. In lower plants, shedding of vegetative parts like gemmae or plantlets help in vegetative reproduction.

15.8 Stress Physiology

Like all other organisms, plants are also subjected to various environmental stresses such as water deficit, drought, cold, heat, salinity and air pollution. The study of functioning of plants under adverse environmental conditions is called **stress physiology**. **Jacob Levitt** (1972) first used the term **biological stress** in relation to plants and according to him stress is "any change in environmental condition that might adversely change the growth and development of a plant".

The reaction of plants facing stress is called **strain**. For example, if a normal plant growing under favourable light conditions is subjected to low light intensity, its photosynthesis is reduced. Thus, low light intensity is referred as stress and reduction of photosynthesis is referred as strain. Biological strains are of two types; Elastic biological strain and Plastic biological strain. If the reaction of plant function is temporary and when it returns to its original state it is called **elastic biological strain**. Example: Temporary wilting. If the reaction is permanent and the plant function does not return to the normal state it is called **plastic biological strain**. Example: Permanent wilting. Some plants get adapted to stress condition and are not adversely affected by stress. Such plants are called **stress resistant** or **stress tolerant** plants. Example: Mangroves. Some plants cannot face stress and they pass their adverse period in dormant state and so they are called **stress enduring plants**. Ephemeral plants are short lived desert plants, which complete their life cycle during the seasonal rains before the onset of dry season. These ephemeral plants are called **stress escapers**. Stress in plants can be classified as given in figure 15.31.

1. Biotic Stresses

These are adverse effects on plants caused by other living organisms such as viruses, bacteria, fungi, parasites, insects, weeds and